

7. Shortest Job First (non-preemptive)

include <stdio.h>

```
int main (){
    int n, i, j;
    int at[10],bt[10],ct[10],wt[10],tat[10],done[10]={0};

    printf("Enter N: ");
    scanf("%d", &n);

    for (i=0;i<n;i++) {
        scanf("%d %d", &at[i],&bt[i]);
    }

    int time=0, count=0;

    while(count<n) {
        int min=999, k=-1;

        for (i=0;i<n;i++) {
            if (at[i]<=time && !done[i] && bt[i]<min){
                min=bt[i]; k=i;
            }
        }

        if (k==-1){
            time++;
            continue;
        }

        time += bt[k];
        ct[k] = time;
        tat[k] = ct[k] - at[k];
        wt[k] = tat[k] - bt[k];

        done[k]=1;
        count++;
    }

    printf("\nP\tAT\tBT\tCT\tWT\tTAT\n");

    for (i=0;i<n;i++) {
        printf("%d\t%d\t%d\t%d\t%d\t%d\n",i,at[i],bt[i],ct[i],wt[i],tat[i]);
    }

    return 0;
```

}